This is **one of the two options** for the second assignment for the Security course at LIACS. You can do this assignment **alone or in pairs**.

Submitting the assignment. In order to submit the assignment, you must first **join an "Assignment 2" group on BrightSpace by Thursday, April 9th 2026 at 23:59h at the latest. These groups are different from the ones from Assignment 1. You must join one of these groups before this date to submit the assignment.** You can find the groups from the course's home page by clicking on "Course tools → Groups". If you do the assignment in pairs, *both members of the group must join the same group.*

Once the deadline for joining groups has passed, you will be added to the assignment on ANS, where you will be able to enter your answers. You must **submit the assignment on ANS by Thursday, April 16th 2026 at 23:59h**. Note that submitting the assignment on ANS means *submitting on behalf of the whole group.* Multiple people can work together on the assignment at the same time (ANS saves progress whenever you navigate away from a question), but the submission is final for everyone.

You must **only submit either this assignment or assignment 2A (the hacker track) but not both**. You are free to try them both out, but only one submission must be made on ANS. Do not even open the assignment that you do not intend to submit on ANS.

1 Motivation

In this assignment you will tackle different problems you can encounter in your career as a Security Manager, e.g., coding, ethics, managing, and general problem solving skills. Please note that some of these questions still contain code; you are studying at LIACS, after all.

2 Task

You are provided with 6 exercises. For each exercise it is indicated how many points can be earned. A maximum of 100 points can be earned in this assignment.

Additional material needed for this assignment can be downloaded from BrightSpace.

For each challenge that you solve you must document your solution. Give complete but concise answers. Some questions have a limit of the number of words/lines you can use. Make sure to follow these instructions.

3 Submission requirements

You will need to enter your answers to each exercise in ANS once submissions open (one week before the deadline). Your answers must be **in English and clear**.

We suggest that you do not wait for submissions to open to start writing your answers, but that you start writing them down in your own notes during the weeks before the deadline.

According to university's regulations, all submissions will be automatically scanned for plagiarism and fraud. Using LLMs is not allowed and will be considered plagiarism. **You must write the answers using your own words**.

Exercise 1: C program [15 pts]

In your work as a Cyber Security Manager, you often need to review code written by your colleagues. Recently, someone pushed the file shown in Listing 1 to a development branch, and is waiting for your approval to merge the code into the production branch. However, something does not look right . . .

```c
#include <stdio.h>
int main(int argc, char** argv) {
        if (argc < 2) {
                return 1;
        }
        char* filename = argv[1];
        // Checks if 1) the file exists and 2) the file only contains allowed commands.
        // Returns false if one of these conditions are not true.
        if (!(exists_and_check_contents(filename))) {
                return 1;
        }

        int option;
        printf("Please give an option:\n 1: Print content\n 2: Execute content");
        scanf("%d", option);

        if (option == 1) {
                // Print the content of the file to the screen
                print_contents_of_file(filename);
        } else if (option == 2) {
                // Execute the command listed in the file
                execute_command_in_file(filename);
        } else {
                printf("Option does not exists.");
        }
        return 0;
}
```

Listing 1: Vulnerable(?) code, written in C

(1a) What is the problem? Why does this happen? Point out the specific lines that contain a problem.

(1b) What is the vulnerability called?

(1c) What can we do to fix the vulnerability? Refer to the line(s) we have to alter.

Exercise 2: Assembly [20 pts]

```
1  main:
2          push    rbp ; --- From here
3          mov     rbp, rsp
4          sub     rsp, 16
5          mov     WORD PTR [rbp-2], 97
6          mov     DWORD PTR [rbp-7], 1953719636
7          mov     BYTE PTR [rbp-3], 0
8          lea     rdx, [rbp-7]
9          lea     rax, [rbp-2]
10         mov     rsi, rdx
11         mov     rdi, rax
12         call    strcpy
13         mov     eax, 0 ; --- To here
14         leave
15         ret
```

Listing 2: Vulnerable(?) assembly code, written for a x86-64 architecture.

Consider the code in Listing 2.

(2a) Explain lines 2 to 13 (included). What do they do? And why? Limit your answer to around 2 lines of explanation per line of code.

(2b) What vulnerability exists in this code?

(2c) How can the vulnerability be fixed?

Exercise 3: Network traffic vulnerabilities [25 pts]

Suppose you are the Security Manager of the networking infrastructure on the Bio Science Park. Just now, a student has come to your office and showed you how they were not able to sign in to BrightSpace using the University WiFi Network, later sending you a screenshot of the login page through e-mail (Figure 1). After submitting their credentials, they are redirected to the same (unsecured) webpage. Luckily, you were just doing a little networking analysis on the network and were able to capture some traffic with Wireshark[1], which you stored inside the file `pcapture.pcap`. Looking at this file, you are getting a little suspicious...

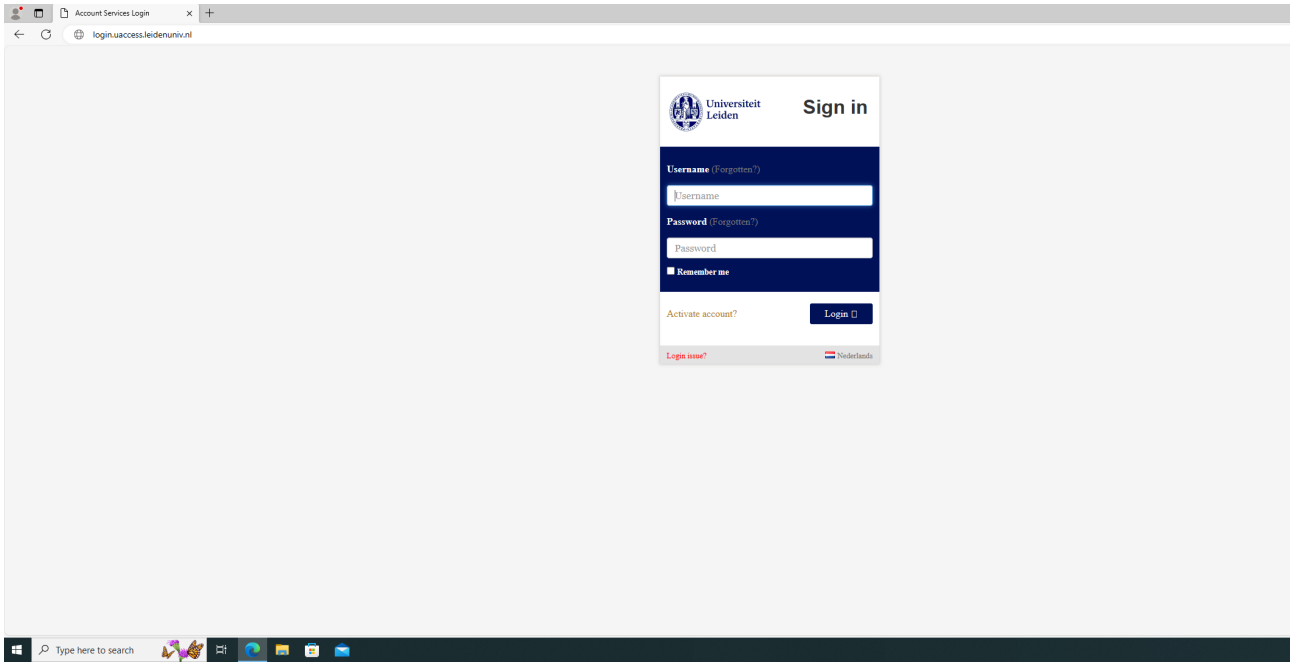

Figure 1: The screenshot the student showed you.

(3a) Take a look at the PCAP file. What do you observe? In your observations, refer to the contents of the file.

(3b) Now, look at the screenshot you were sent by the student (Figure 1). What do you conclude?

(3c) What are your next steps (if any) for prevention?

Exercise 4: Privacy laws [17 points]

On May 1, 2018, the Dutch 'Intelligence and Security Services Act 2017' entered into force. This act has the (not so positive) nickname as 'De Sleepwet' ('The Draglaw'). The following 3 questions are about this law.

(4a) How did the act obtain this nickname?

(4b) Give your thoughts about this act. Do this in a long-form answer of around 150-200 words, where you give at least two arguments which support your opinion.

(4c) The Dutch Intelligence Service (AIVD) warns organisations that enemy actors are eavesdropping on their communications. However, all this communication is encrypted, and it will take a lot of time to decrypt this data. We can assume that the organisations do not send any not-encrypted data by accident. Why would the enemy actors still eavesdrop the communication? And which countermeasure would make the eavesdropping more useless?

[1] `https://www.wireshark.org/`

As a Manager, you should be familiar with the several General Data Protection Regulations in your country. The following 2 questions are about the GDPR. You can assume that we want the information to comply with the European Union GDPR[2].

(4d) The data of your site is breached, and an ethical white-hat hacker has made you aware of this. How fast should you inform the national Data Protection Authority?

(4e) Name 3 ways of how data could get leaked and give countermeasures to mitigate it.

Exercise 5: Encryption and hashing [13 pts]

Imagine that you are running a website where users can register and log-in. Using this site, users can see the home addresses of other users, which can then be used to send postcards to each other. Since you are the Security Manager of this site, you should make sure that all the data is stored securely (with respect to confidentiality, integrity and availability).

The site asks the following data from the users:

- Username
- Full name
- Mail
- Password
- Home address

(5a) Create a small database schema of the site.

(5b) Which data should be encrypted? And which data should be hashed? Why?

(5c) For both hashing and encryption name which algorithms you would use, and which ones you would definitely avoid. Name at least one algorithm for each of these four categories, and justify your choices.

Exercise 6: Authentication riddle [10 pts]

You are given on BrightSpace a vulnerable Python file named `encryption.py`.

(6a) What is happening here? What can be exploited?

(6b) How can you exploit this weakness?

(6c) Find the secret key, and show your steps.

[2] `https://europa.eu/youreurope/business/dealing-with-customers/data-protection/data-protection-gdpr/index_en.htm`